INORGANIC CUMULATIVE EXAM

October 26, 1996

Part I (any 4)

Give a reasonable synthesis for each of the following starting from readily available materials. Be sure to indicate any special conditions, e.g. temperature, pressure, catalyst etc., that are required.

a) NH_3

b) cis -$Pt(C_5H_5N)_2Cl_2$

c) $[PN(Cl)_2]_3$ hint: Cl bonded to P

d) ICl

e) $CH_3Mn(CO)_5$ from $Mn_2(CO)_{10}$

Part II (any 4)

For each of the common reagents below, give the formula, preparation, structure, and an example of a reaction that illustrates their reactivity or use as indicated.

a) Vaska's compound [hint: Ir(I)] - formula/structure/reactivity

b) Zeise's salt - preparation/structure

c) aqua regia - preparation/reactivity

d) solvated electron - preparation/reactivity

e) NaK alloy - preparation/reactivity

Part III (any 4)

It is often necessary to generate simple gaseous reagents in the laboratory from readily available and common materials. Give a reaction that indicates how you would generate each of the following, include any special conditions.

a) HCl

b) CO_2

c) Cl_2

d) N_2

e) O_3